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ALIAH UNIVERSITY

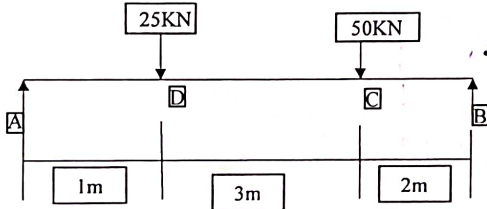
Department of Civil Engineering
End-Semester Examination- January, 2024

Year: 2nd
Subject: Mechanics of Materials
Full Marks: 80

Semester: 3rd
Subject Code: CENUGPCO1
Time: 3 hrs.

Attempt Any Five questions and assume suitable data if required.
All Questions are of Equal Marks.

1. Draw Shear Force and Bending Moment Diagram for the structure as shown below



$$+ (37.5 \times 6) - (50 \times 4) - (25 \times 1)$$

225 -

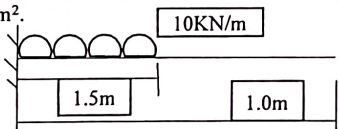
$$\frac{8 \text{ m}}{EI \text{ N/mm}^2}$$

$$\left(\frac{8}{EI} \right) \times 10^3$$

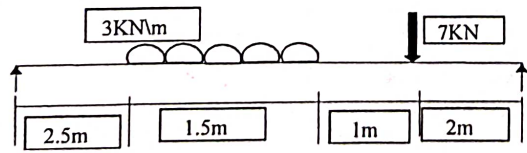
$$\frac{8}{1.9 \times 10^{12}} \times 10^3$$

10.9 mm.

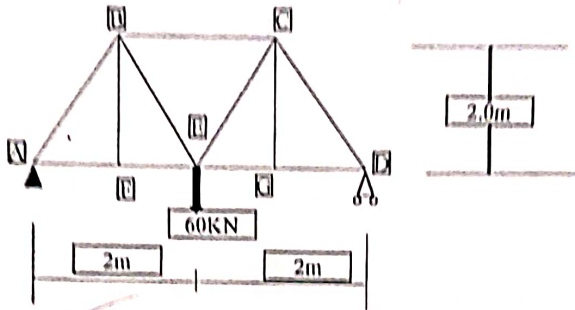
2. Calculate the slope and deflection at the free end of the cantilever.
Take $EI = 1.9 \times 10^{12} \text{ N-mm}^2$.



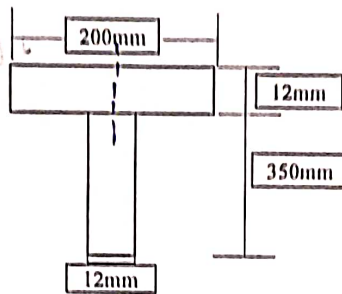
3. Draw Shear Force and Bending Moment Diagram for the structure as shown below.



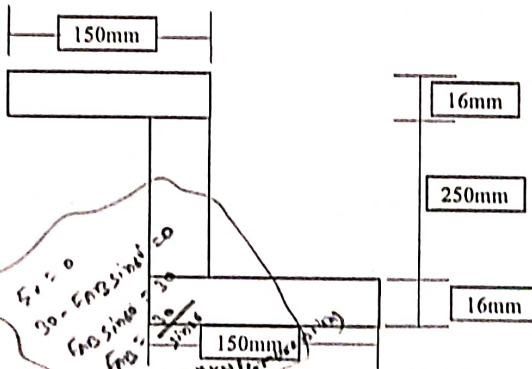
4. Analyse the Truss as drawn below



5. Calculate the Centre of Gravity (CG) and Moment of Inertia (MI) about the CG for the section as shown below.



6. Calculate the Centre of Gravity (CG) and Moment of Inertia (MI) about the CG for the section as shown below.



$\Sigma F_x = 0$
 $R_A + R_D = 0$
 $\Sigma F_y = 0$
 $R_A + R_D - (60 \times 2) = 0$
 $R_A = 60$
 $R_D = 60$

Consider Joint A,

$\Sigma H = 0$
 $F_{AE} - F_{AB} \cos 60^\circ = 0$
 $F_{AE} = F_{AB} \cos 60^\circ$
 $= 1.15 \text{ (kN) Tension}$

Consider Joint D

$\Sigma H = 0$
 $F_{DE} - F_{DC} \cos 60^\circ = 0$
 $F_{DE} = F_{DC} \cos 60^\circ$
 $= 1.15 \text{ kN (Tension)}$

$\Sigma F_x = 0$
 $30 - F_{ED} \sin 60^\circ = 0$
 $30 - F_{ED} \sin 60^\circ = 0$
 $F_{ED} = 34.64 \text{ kN}$

$\Sigma F_y = 0$
 $30 - F_{ED} \cos 60^\circ = 0$
 $30 - F_{ED} \cos 60^\circ = 0$
 $F_{ED} = 34.64 \text{ kN (compression)}$
 $AE = 1.15 \text{ Tension}$
 $AB = 34.64 \text{ Compression}$
 $DE = 1.15 \text{ Tension}$
 $CD = 34.64 \text{ Compression}$

ALIAH UNIVERSITY
ODD SEMESTER EXAMINATION, AUTUMN-2024
PAPER: MATUGBS03 (Engineering Mathematics III)

Full Marks: 80, Time: 3 Hours

Use different answer scripts for each unit, All symbols have their usual meanings

Unit-I

Answer any **EIGHT** from the following questions:

$8 \times 5 = 40$

1. Solve the following partial differential equation by Lagrange's method 5

$$z(x+y)p + z(x-y)q = x^2 + y^2.$$

2. Solve the following partial differential equation by Lagrange's method 5

$$(3z - 5y)p + (5x - 4z)q = 4y - 3x.$$

3. Solve the following non-linear partial differential equation by Charpit's method 5

$$z^2(p^2 z^2 + q^2) = 1.$$

4. Solve the following non-linear partial differential equation by Charpit's method 5

$$2zx - px^2 - 2qxy + pq = 0.$$

5. Eliminate the constants a and b to form the PDE 3+2

$$(a)z = (x - a)^2 + (y - b)^2; \quad (b)z = ax + by + ab.$$

6. Eliminate the function ϕ to form the PDE 5

$$\phi(x + y + z, x^2 + y^2 + z^2) = 0.$$

7. Classify the following PDEs 2+3

(a) $4u_{xx} + 6u_{xy} + 9u_{yy} + 100u_x - 100u_y = 0,$

(b) $2u_{xx} + 4u_{xy} + 3u_{yy} - 10u_x - 7u_y = 2.$

8. Find the inverse of the Laplace transformation 5

$$L^{-1} \left[\frac{p^2 - 3p + 4}{p^3} \right].$$

9. (i) Find degree and order of the PDE 2+3

$$\frac{\partial^2 z}{\partial x^2} = \left(1 + \frac{\partial z}{\partial y} \right)^{\frac{3}{2}}.$$

- (ii) Form the partial differential equation by eliminating the function f from the equation

$$y = f(x - at) + f(x + at).$$

10. Find the Laplace transformation of the following functions

5

$$(a) f(t) = e^{-3t} \cos 3t, \quad (b) g(t) = \sin^2 t - t.$$

11. Find the Fourier series for $f(x) = |\cos x|$ in $(-\pi, \pi)$ of periodicity 2π .

5

12. Derive the Fourier series of $f(x) = x + x^3$ in $(-\pi, \pi)$ of periodicity 2π and hence deduce $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

5

Unit-II

Answer any EIGHT from the following questions:

8 × 5 = 40

1. What does the equation

$$x^2 + 4xy + y^2 - 2x + 2y + 6 = 0$$

become when origin moved to the point $(-1, 1)$ and the axes are turned through an angle of 45° in anticlockwise direction.

5

2. Prove that the equation to the straight lines through the origin each of which makes an angle α with the straight line $y = x$ is $x^2 - 2xy \sec 2\alpha + y^2 = 0$.

5

3. Prove that the equation $x^2 + 6xy + 9y^2 + 3x + 9y = 10$ represents a pair of parallel straight lines and find the distance between them.

5

4. Prove that the straight line $px + qy + r = 0$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $r^2 = a^2 p^2 + b^2 q^2$.

5

5. If the pole of the straight line with respect to the circle $x^2 + y^2 = c^2$ lies on $x^2 + y^2 = k^2 a^2$, then prove that the straight line will touch the circle $x^2 + y^2 = a^2$.

5

6. Reduce the following equations to their canonical forms and determine the nature of the conic represented by the equation $3x^2 + 2xy + 3y^2 - 16x + 20 = 0$.

5

7. Show that the pair of straight lines whose direction cosines are given by $3lm - 4ln + mn = 0$ and $l + 2m + 3n = 0$ are at right angles.

5

8. (a) Find the equation of the plane parallel to the plane $2x - 2y - z = 3$ and situated at a distance 7 unit from it.

3

(b) If a straight line makes an angle $\frac{\pi}{3}$ with x -axis, $\frac{\pi}{4}$ with y -axis and $\theta \in [0, \pi]$ with z -axis. Then find the value of θ .

2

9. Find the equation of the plane which passes through the point $(2, 1, -1)$ and orthogonal to each of the planes $x - y + z = 1$ and $3x + 4y - 2z = 0$.

5

10. Find the image of the point $(-3, 8, 4)$ in the plane $6x - 3y - 2z + 1 = 0$.

5

11. Find the shortest distance between the straight lines

5

$$\frac{x}{1} = \frac{y-1}{0} = \frac{z-1}{1} \quad \text{and} \quad \frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}.$$

12. Find the centre and the radius of the circle

5

$$x^2 + y^2 + z^2 = 25, x + 2y + 2z + 12 = 0.$$

ALIAH UNIVERSITY

B. Tech. Semester:III, Civil Engineering

Odd Semester Examination Jan-2024

Subject Code & Name: (CENUGPC03), Building Materials

Time:3 Hours

Total Marks:80

- Instruction: (1) Figures to the right indicate full marks.
(2) Assume suitable data if required.
(3) Question-1 is compulsory.

Que: -1 Answer the following questions

2×10=20

- ☒ A Describe the different varieties of lime ✓
- ☒ B What is efflorescence in bricks? What are its causes and remedies. ✓
- ☒ C What is the use of Vicat's apparatus. Draw the neat sketch. ✓
- ☒ D How the entrapped air plays a role in concrete strength. Describe in graphical form. ✓
- ☒ E Write a note on Stretcher and Header bond with neat sketch. ✓
- ☒ F What are the precautions to be exercised while using mortars? ✓
- ☒ G Define terms- Flight, Landing, Soffit, Nosing. ✓
- ☒ H What is bulking of sand? How does it affect concrete mix? ✓
- ☒ I Differentiate between: Star shake and heart shake. ✓
- ☒ J What should be the slump height and compaction factor of- Precast slab, Road works. ✓

Que: -2 Attempt any Six.

10×6=60

- ☒ A Draw the cross-section of a matured tree. Discuss each part in details. ✓
- ☒ B How do you determine the Flakiness Index and Elongated Index? What is its significance? ✓
- ☒ C Enlist and explain the different types of Admixtures used in Concrete. ✓
- ☒ D Describe compaction factor test in detail with neat sketch. ✓
- ☒ E State the chief impurities in limestones? How do they affect the properties of lime? ✓
- ☒ F What are the initial and final setting times of cement? What is their importance? How to determine setting time in laboratory? ✓
- ☒ G Describe briefly- Cement paint, Oil paint, French paint, Enamel, Wax polish. ✓

Aliah University
Theoretical Examination, 2024
UG section (Science & Engg/Autumn Semester)
Sub: Indian Constitution (UCCUGAU03)

Full Marks: 80

Time: 3hrs

1. Write any ten questions: (within 30 words each) 2x10=20
- a. Which principles were added to the Constitution of India by the 42nd Amendment Act?
 - b. Discuss Article 17 of the Constitution of India.
 - c. Discuss Article 21A of the Constitution of India. ✓
 - d. Explain the term of 'Prohibition'.
 - e. Write down any two fundamental duties mentioned in our Constitution.
 - f. What is the composition of the Electoral College that elects the President?
 - g. What is 'Ordinance'? ✓
 - h. What do you mean by the impeachment of the President of India?
 - i. Discuss any two functions of the Speaker of the Lok Sabha.
 - j. Mention the qualifications of the members of the Lok Sabha.
 - k. Discuss the composition of the Upper House of the Indian Parliament.
 - l. Mention two areas where the Rajya Sabha is more powerful than the Lok Sabha.
1. Write any four questions: (within 100 words each) 5x4=20
- a. Write a short note on the writs of Habeas Corpus and Quo Warranto. ✓
 - b. Critically discuss the Right to Freedom of Religion. ✓
 - c. Enumerate the differences between the Fundamental Rights and the Directive Principles of State Policy.
 - d. Describe the election procedure for the President of India.
 - e. Mention Five Fundamental Duties mentioned in our Constitution. ✓
 - f. Write any five Directive Principles of the State Policy.
2. Write any four questions: (Within 350 words each) 10x4= 40
- a. Discuss the basic features of the Indian Constitution. ✓
 - b. Discuss articles 14 to 18 of the Indian Constitution. ✓
 - c. Discuss Article 19 as enumerated in the Constitution of India and point out its reasonable restrictions.
 - d. "The Constitution of India is neither purely federal nor unitary but a combination of both". Do you agree with this view? Give reasons in support of your answer. ✓
 - e. Discuss the powers and functions of the Prime Minister of India. ✓
 - f. Analyse the relationship between the two houses of Parliament.

(C/A)

17 → 100%
18 → 100%

Odd Semester Examination 2023-24

Subject Code: CENUGPC02

Subject Name: Surveying and Geomatics

B. Tech Civil Engineering IIISemester

Full Marks: 80 (8x10)

Time: 3 Hrs.

Attempt Any 8 Question

1. The plan of an area drawn with an original scale of $1\text{cm}=10\text{m}$, has shrunk that a line, originally 25cm long on the plan, now measures 24.5cm . Find the length of shrunk scale given by 1cm ? Describe the principle of survey. ✓
2. Explain the significance of Electromagnetic Radiation (EMR) in remote sensing with suitable diagram? ✓
3. What do you mean by triangulation? Describe well and ill-conditioned triangle? ✓
4. Elaborate the different types of surveys based on the instrument and nature of the survey? ✓
5. Convert the following quadrantal bearing into whole circle bearing:
(a) $N25^{\circ}30'E$ (b) $S50^{\circ}E$ (c) $S25^{\circ}45'W$ (d) $N15^{\circ}W$ ✓
6. Define the following terms:
(a) Isogonic Line (b) Aclinic Line (c) Agonic Line (d) Isoclinic Line
7. What are the working principles of the Global Navigation Satellite System (GNSS) and Global Positioning System (GPS) with a suitable diagram? ✓
8. Explain the concept of geospatial technology, and how are they used in sustainable natural resource management?
9. Differentiate between raster and vector datasets with example, and also write down the advantages and disadvantages. ✓

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Odd (3rd) Semester Examination, 2023-24

Course: Civil Engineering

Year: Second Year

Subject: Introduction to Civil Engineering

Subject Code: CENUGHUOI

Max. Marks: 80

Time: 3 hrs

1. Answer any four questions:

(20X4=80)

a) What are the different types of foundations normally used in the construction sector. Briefly explain them as per their respective applicability to the type of construction.

(5+15=20)

b) Differentiate between concrete and mortar. What do you mean by water cement ratio. What is meant by characteristic compressive strength of concrete. What are the common types of bricks used in brickwork?

(5+5+5+5=20)

c) What are the considerations for selecting the tread and rise of a staircase? Briefly describe a dog legged staircase. What is meant by fineness, crushing strength, soundness and final setting time of cement? Differentiate between mild steel and high carbon steel. What is meant by curing of concrete? Differentiate between artificial and natural seasoning of timber.

(3+3+6+3+3+2=20)

d) Briefly explain the general water supply scheme of an area through a flow chart. How is the total water quantity requirement for an area computed for preparing a water supply scheme of the same? What are meant by "Green Buildings"? What are the different ratings of "Green Buildings" as per the certifying authorities? Explain the importance of the selection of materials and orientation in designing a green building.

(5+4+3+4+4=20)

e) Explain the utmost importance of IS Codes in the discipline of Civil Engineering with reference to three relevant codes pertinent to three streams of Civil Engineering. Name some structures of civil engineering brilliance in and outside India. What is meant by the term superelevation related to highway engineering? What are the different types of sewers normally encountered? Differentiate between plinth level and ground level of a building.

(5+4+3+4+4=20)